BASE CHANGE FOR A PUSHFORWARD OF A SHEAF OF SETS ALONG A FLAT MORPHISM WITH GEOMETRICALLY CONNECTED STRICT HENSELISATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to stalks via the stalk criterion, the two displayed computations of the stalks as sections over fibre products, and the identification through sections upstairs applied to the base change morphism.

Editorial note. The label lemma-base-change-f-star names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement's cartesian diagram is the author's and is delivered verbatim; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Equality of sheaves can be checked on stalks. Same chapter, the theorem stating that a map of etale sheaves is an isomorphism if and only if it is so on geometric stalks. Supplies the reduction of the identity of sheaves to a comparison of stalks.
- Stalks of pushforwards as sections over a fibre product. Same chapter, the theorem describing the stalk of a pushforward as the sections over the corresponding fibre product. Supplies the two displayed identities of stalks.
- Sections upstairs for a morphism with geometrically connected fibres. Same chapter, the lemma comparing sections of a pullback with sections of the original sheaf for such a morphism. Supplies the equality of the two section sets, applied to the base change morphism of strict henselisations.

Source reference. Original reference 0A3H: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5030–5030. Prerequisite pointer invoked at proof line(s) 16820; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03PU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3546–3546. Prerequisite pointer invoked at proof line(s) 16808; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 03Q9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8015–8015. Prerequisite pointer invoked at proof line(s) 16809; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to S$ be a flat morphism of schemes such that for every geometric point $\overline{x}$ of X the map*

$$\mathcal{O}^{sh}_{S,f(\overline{x})} \longrightarrow \mathcal{O}^{sh}_{X,\overline{x}}$$

has geometrically connected fibres. Then for every cartesian diagram of schemes

$$\begin{array}{ccc} X & \xleftarrow{h} & Y \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle e} \\ S & \xleftarrow{g} & T \end{array}$$

with g quasi-compact and quasi-separated we have $f^{-1}g_*\mathcal{F} = h_*e^{-1}\mathcal{F}$ for any sheaf $\mathcal{F}$ of sets on $T_{étale}$.

Proof. It suffices to check equality on stalks, see Theorem 03PU. By Theorem 03Q9 we have

$$(h_*e^{-1}\mathcal{F})_{\overline{x}} = \Gamma(\mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) \times_X Y, e^{-1}\mathcal{F})$$

and we have similarly

$$(f^{-1}g_*^{-1}\mathcal{F})_{\overline{x}} = (g_*^{-1}\mathcal{F})_{f(\overline{x})} = \Gamma(\mathrm{Spec}(\mathcal{O}^{sh}_{S,f(\overline{x})}) \times_S T, \mathcal{F})$$

These sets are equal by an application of Lemma 0A3H to the morphism

$$\mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) \times_X Y \longrightarrow \mathrm{Spec}(\mathcal{O}^{sh}_{S,f(\overline{x})}) \times_S T$$

which is a base change of $\mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) \to \mathrm{Spec}(\mathcal{O}^{sh}_{S,f(\overline{x})})$ because $Y = X \times_S T$. $\square$